

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element L Initial Template AT - 6 points

Element L: Presentation of Designer's Recommendations

Content of Recommendations (L1, L2)

- What could be done next to go further with this project?

If I were to go further, I would continue trying to make a sprinkler that is able to spread water or research the conditions in the greenhouse more and see if there are any previously unforeseen problems that might have needed to be accommodated for. I could also test out a completely different direction, for example, a bottom watering system instead of an overhead sprinkler system.

- What would you keep if you did the project again?

If I were to repeat this project I would keep the angled pipe as that part was the most successful piece of the project, being able to prevent mold and distribute water to its holes/sprinklers. In addition, I think that it would be helpful to keep the testing plans, specifically the mold-less test and the coverage test. The mold-less testing plans were able to accurately simulate real-use while the coverage test plans were detailed and outlined a specific criteria that helped me figure out what/where to aim for each consecutive iteration on the sprinklers.

- What would you change in your current prototype or testing plan?

The biggest failure of the design was with the sprinklers, wherein none of the sprinkler models made were able to successfully distribute water far enough. If I were to re-attempt this, I would first start by researching DIY sprinklers that people have made for similar setups as well as learn more about creating pressure. Then I would have drafted some basic ideas based on what I learned, or just follow a youtube tutorial,

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depending on the materials available or quality of the brainstormed ideas, instead of mindlessly throwing out and testing ideas until I burned myself out. Finally I would test and iterate like before, making sure to test under conditions that would be most similar to the actual final plan.

- What would have made this project easier for you to complete?

The biggest factor that would have made this project easier to complete would be to have reliable teammates that contributed consistently and put in good effort, as it would have made the workload much more manageable and a lot less stressful. On the topic of ideas, later on in the process, I consistently ran into problems in which I realized we had overlooked important parts of the prototype that then needed to be improvised, which further caused their own issues that had to be trialed over and over. It would have also been helpful if I were more thorough with doing research at the beginning, specifically into sprinklers as figuring out the most complicated and essential part then working from that would have been much easier than the reverse, which is what actually happened. Finally, having a bigger budget and access to more materials, especially time, would have been helpful as it would have helped to get out better or more prototypes and allowed for more tests to happen.

Step L: Presentation of Designer's Recommendations

	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
L1. Content of recommendations. Recommendations include _____.	specific and detailed ways the project can be improved and a rationale for why they are needed	specific and detailed ways the project can be improved	some specific ways the project can be improved	one way the project can be improved	vague recommendation(s) for improvement of the project	no recommendations for improvement of the project



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	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
L2. Detailed plans for implementation of improvements. Plans for implementation of improvements are _____.	consistently detailed	generally detailed	minimally detailed and may include one or more caveats	partially detailed and/or overly general	vague/unclear or minimally related to the recommendations provided	not offered